

EIGHT CONDITIONS ON THE MISSING ESTIMATE FOR THE THREE-DIMENSIONAL NAVIER–STOKES EQUATIONS

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ABSTRACT. Every known route to global regularity for the three-dimensional incompressible Navier–Stokes equations stops at the same place: the only unconditional bound is the energy inequality, which is supercritical by one half of a derivative, and no scale-invariant quantity is known to be bounded a priori. This note does not close that gap. It assembles, from results that are mostly not ours, a list of eight conditions that any quantity deciding the problem must satisfy, each tied to a theorem or a construction that rules out the alternative. Two of the conditions carry new content. The first is a parity condition: Navier–Stokes preserves mirror symmetry and helicity is odd under reflection, so helicity, and every pseudoscalar invariant, vanishes identically on an invariant class in which the known blowup constructions for model equations are carried out; the Fourier-restricted model of Miller, which keeps the energy equality and the exact enstrophy-growth identity and blows up, has helicity identically zero. The second is a transport condition, read off from Miller’s one-parameter family of strain models, all sharing the enstrophy identity and constraint space of the true equation: the family is regular at one end and blows up in its interior, which places the burden of any regularity proof on advection. Along the way we record an elementary criterion: a smooth solution stays regular as long as a single helical component lies in the scale-critical class $L_t^\infty \dot{H}^{1/2} \cap L_t^2 \dot{H}^{3/2}$. We also note that the geometric conclusion of the middle-eigenvalue criterion of Neustupa–Penel and Miller, that blowup must be driven by planar stretching against axial compression, agrees with the structures singled out by several independent programs.

1. INTRODUCTION

We write the incompressible Navier–Stokes equations on $\mathbb{R}^3$ or $\mathbb{T}^3$ as

$$\partial_t u - \nu \Delta u + (u \cdot \nabla)u + \nabla p = 0, \quad \nabla \cdot u = 0, \quad (1)$$

with vorticity $\omega = \operatorname{curl} u$ and strain $S = \frac{1}{2}(\nabla u + \nabla u^T)$. The equations are invariant under $u_\lambda(x, t) = \lambda u(\lambda x, \lambda^2 t)$. We say a quantity has dimension L^a when it scales as λ^{-a} under this map with ν fixed; a quantity is critical when $a = 0$.

The energy inequality bounds $\sup_t \|u\|_{L^2}^2 + 2\nu \int \|\nabla u\|_{L^2}^2$ by the initial energy. The norm $\|u\|_{L^2}$ has dimension $L^{1/2}$. Every known regularity criterion is stated in a critical quantity: $L_t^p L_x^q$ with $2/p + 3/q = 1$ [27], the endpoint $L_t^\infty L_x^3$ [11], $\int \|\omega\|_\infty dt$ [3], $\dot{H}^{1/2}$ through the Fujita–Kato theory [12]. The gap between the two is half a derivative, and it has not moved since these results were proved.

The purpose of this note is bookkeeping. Over the last two decades a series of results, several of them constructions of blowup for model equations, has made it possible to say with some precision what a quantity closing the gap would have to look like. Each such result excludes a class of candidates or a class of arguments. We collect them as eight conditions (§3), give short proofs of the three statements that appear to be new (§4, §5), and record two observations that follow from reading the programs side by side (§6, §8). Section 9 tests the known candidates against the list.

The conditions are of two kinds. Some are theorems: a candidate violating them is refuted by an explicit solution or construction. Others are inferences from model equations, which constrain arguments rather than quantities and could in principle be circumvented. We mark which is which. Nothing here proves or disproves regularity.

2. THE SETTING

Let Φ be a functional on smooth divergence-free fields. We say that Φ *decides regularity* if

- (i) (a priori) along every smooth solution of (1) on $[0, T)$, $\sup_{t < T} \Phi(u(t)) \leq F(\Phi(u_0), \|u_0\|_{L^2})$ for some function F independent of T ; and
- (ii) (control) if $\sup_{t < T} \Phi(u(t)) < \infty$ then u extends smoothly past T .

Energy satisfies (i) and not (ii). Every critical norm satisfies (ii) and is not known to satisfy (i). The question is whether anything satisfies both.

3. EIGHT CONDITIONS

Condition C1 (scaling). Φ is invariant under $u \mapsto u_\lambda$, or controls a quantity that is.

This is the familiar requirement. A supercritical quantity of dimension L^a with $a > 0$ can stay bounded while u_λ concentrates, because it decays under the rescaling that concentrates the solution. It is also the form in which every known control statement (ii) is proved.

Condition C2 (a priori). The bound (i) holds for all smooth data.

No critical quantity is known to satisfy C2 in general. The special classes where one is known are informative. For axisymmetric flow the swirl $\Gamma = ru^\theta$ obeys a maximum principle and is critical, which underlies the exclusion of Type I blowup [15, 7]; for stationary flow the head pressure $Q = \frac{1}{2}|u|^2 + p$ satisfies $\Delta Q - u \cdot \nabla Q = |\omega|^2 \geq 0$ and obeys a maximum principle [5]. Koch, Nadirashvili, Seregin and Šverák note that for the general three-dimensional problem “there is no known similar quantity” [15], and this remains the case.

Condition C3 (coercivity). Φ controls a critical norm, for instance $\|u\|_{\dot{H}^{1/2}}$ or $\|u\|_{L^3}$, so that (ii) follows from the Fujita–Kato theory or from [11].

Conserved or monotone quantities that are sign-indefinite fail C3; helicity is the example discussed in §5.

Condition C4 (structure). The proof of (i) does not apply to Tao’s averaged Navier–Stokes equation [29], nor to Euler–Arnold equations with inhomogeneous inertia operators within ε of the Euler metric [30].

This is a theorem. The averaged equation shares the energy identity and the harmonic-analysis estimates of (1) and blows up [29]; the Lagrangian modifications of [30] share, in addition, the Lie–Poisson structure, Kelvin’s theorem, helicity and vorticity transport, and blow up. An argument using only these ingredients is refuted. The structure used must be exact. Momentum helicity is a quantity that the cascade operator of [29] does not conserve for any positive definite inertia operator [32], so a helicity-based argument is not refuted by [29]; it fails for other reasons (§5).

Condition C5 (viscosity). If Φ is also bounded a priori along solutions of the Euler equations, then the passage (ii) from bounded Φ to regularity uses $\nu > 0$ in an essential way.

Euler solutions from $C^{1,\alpha}$ data on $\mathbb{R}^3$ [10], and from smooth data in a cylinder with a wall [6], blow up in finite time. A mechanism that would also prevent those singularities is refuted. Condition C5 is compatible with conserved quantities at the level of $\dot{H}^{1/2}$: bounds in $\dot{H}^{1/2}$ suffice for (1) and not for Euler. A claimed construction of blowup for the unforced Euler equations on $\mathbb{R}^3$ from smooth data [22] is not independently verified at the time of writing; if it is correct it strengthens C5 without changing it.

Condition C6 (limits). Φ is lower semicontinuous under local convergence and invariant under translation and scaling, so that its bound passes to blowup limits and to ancient solutions.

This is what the Liouville approach requires [15]: zooming in at a putative singularity produces a bounded ancient solution, and the argument closes if the limit inherits a bound that forces it to be constant. The swirl Γ is the model case, since $\Gamma_\lambda(x) = \Gamma(\lambda x)$.

Condition C7 (parity). Φ is not odd under reflections. More precisely, Φ must be informative on the class of mirror-symmetric solutions, which (1) preserves.

This is a theorem, proved in §4: every pseudoscalar functional vanishes identically on the mirror-symmetric class, and that class contains the settings of the known blowup constructions for model equations.

Condition C8 (transport). The proof of (i) uses the advection term, and not only the enstrophy identity, the constraint that S is the strain of a divergence-free field, and the scaling.

This is an inference from Miller's strain models, explained in §7. It constrains arguments, not quantities, and it is the least rigid of the eight.

4. THE PARITY CONDITION

Let R be an orthogonal matrix with $\det R = -1$ (on $\mathbb{T}^3$, one preserving the lattice, such as a coordinate reflection), and set $(P_R u)(x) = Ru(R^{-1}x)$, $(P_R p)(x) = p(R^{-1}x)$. Since (1) is invariant under the full orthogonal group acting in this way, $P_R u$ solves (1) whenever u does.

Proposition 4.1. *Let $H(u) = \int u \cdot \omega \, dx$. Then $H(P_R u) = -H(u)$. If $u_0 = P_R u_0$ then the smooth solution satisfies $u(t) = P_R u(t)$ for all t in its lifespan, and $H(u(t)) = 0$ identically. The same holds for every functional Φ with $\Phi(P_R u) = -\Phi(u)$.*

Proof. For $\det R = -1$ one has $\text{curl}(P_R u) = -P_R(\text{curl } u)$ as vector fields, because the curl is a pseudovector: R^{-T} acting on the ε tensor produces the factor $\det R$. Hence $(P_R u) \cdot \text{curl}(P_R u)(x) = -(u \cdot \omega)(R^{-1}x)$, and integrating with $|\det R| = 1$ gives $H(P_R u) = -H(u)$. If $u_0 = P_R u_0$ then u and $P_R u$ are smooth solutions with the same data, so they coincide by uniqueness. Then $H(u(t)) = H(P_R u(t)) = -H(u(t))$. The last sentence is the same argument. $\square$

On the invariant class the two helical components of u , defined in §5, are exchanged by the reflection and have equal norms in every $\dot{H}^s$, so helicity carries no information there at all.

The invariant class is not exotic. Miller's Fourier-restricted Euler and hypodissipative Navier–Stokes equations [18] keep the energy equality and the exact identity for enstrophy growth of the true equations, and blow up in finite time for the inviscid case and for dissipation $(-\Delta)^\alpha$ with $\alpha < \log 3/(6 \log 2)$; his blowup ansatz is mirror symmetric. More is true of that model.

Lemma 4.2. *Every field in Miller's constraint space has helicity zero.*

Proof. In [18] the Fourier coefficient at each admissible frequency k is restricted to $\hat{u}(k) = a_k v_k$, where $v_k = P_k^\perp \sigma / |P_k^\perp \sigma|$ is a real unit vector orthogonal to k , $\sigma = (1, 1, 1)$, and $v_{-k} = v_k$. Then $\hat{\omega}(k) = i a_k k \times v_k$ and $\hat{u}(k) \cdot \overline{\hat{\omega}(k)} = -i |a_k|^2 v_k \cdot (k \times v_k) = 0$ for every k . Summing over k gives $H = 0$. $\square$

Each mode is linearly polarized, an equal mixture of the two helicities. So a model that respects the energy equality, the enstrophy identity, and (trivially) helicity conservation still blows up. The claimed unforced Euler construction of [22] uses odd data, $u_0(-x) = -u_0(x)$, which is invariance under the point reflection $R = -I$; there too the helicity vanishes. We conclude that no pseudoscalar quantity satisfies both C3 and C2.

The companion result [32] is unaffected: it classifies operators, not solutions, and uses momentum helicity as an identity that a class of operators must satisfy.

5. HELICITY: THE NEAR MISS

Helicity is worth recording because it satisfies every condition except C3 and C7. On $\mathbb{T}^3$, write

$$\hat{u}(k) = a_+(k)h_+(k) + a_-(k)h_-(k), \quad ik \times h_\pm(k) = \pm |k|h_\pm(k),$$

with $h_\pm(k)$ orthonormal eigenvectors of the curl on $k^\perp$ [9, 31], and define the helical components $u_\pm = \sum_k a_\pm(k)h_\pm(k)e^{ik \cdot x}$. Then

$$H = \sum_k |k|(|a_+|^2 - |a_-|^2), \quad \|u\|_{\dot{H}^{1/2}}^2 = \sum_k |k|(|a_+|^2 + |a_-|^2), \quad (2)$$

so helicity is critical and lives exactly at the level of the missing half-derivative; it fails C3 only through the minus sign. Along a smooth solution of (1),

$$\frac{d}{dt}H = -2\nu \int \omega \cdot \operatorname{curl} \omega \, dx = -2\nu (\|u_+\|_{\dot{H}^{3/2}}^2 - \|u_-\|_{\dot{H}^{3/2}}^2). \quad (3)$$

When the sign is fixed the problem closes. Biferale and Titi [4] restrict the dynamics to one helical sign, where H is equivalent to $\|u\|_{\dot{H}^{1/2}}^2$, and obtain global regularity; they suggest that singular solutions of (1), if any, should be sought among the triad interactions that mix helicities. The same identities give a criterion for the full equation.

Proposition 5.1. *Let u be a smooth solution of (1) on $\mathbb{R}^3 \times [0, T)$ with $u_0 \in \dot{H}^{1/2}$, the helical decomposition being taken with the Fourier transform. If*

$$u_- \in L^\infty(0, T; \dot{H}^{1/2}) \cap L^2(0, T; \dot{H}^{3/2}),$$

then u extends smoothly past T . The same holds with the roles of u_+ and u_- exchanged.

Proof. Integrating (3),

$$H(t) + 2\nu \int_0^t \|u_+\|_{\dot{H}^{3/2}}^2 = H(0) + 2\nu \int_0^t \|u_-\|_{\dot{H}^{3/2}}^2.$$

By (2), $\|u_+(t)\|_{\dot{H}^{1/2}}^2 = H(t) + \|u_-(t)\|_{\dot{H}^{1/2}}^2$, so

$$\|u_+(t)\|_{\dot{H}^{1/2}}^2 + 2\nu \int_0^t \|u_+\|_{\dot{H}^{3/2}}^2 \leq H(0) + 2\nu \int_0^t \|u_-\|_{\dot{H}^{3/2}}^2 + \sup_{s \leq T} \|u_-(s)\|_{\dot{H}^{1/2}}^2.$$

The right side is finite. Hence $u \in L^\infty(0, T; \dot{H}^{1/2}) \subset L^\infty(0, T; L^3)$, and the solution does not blow up at T by the endpoint criterion of Escauriaza, Seregin and Šverák [11]. The identities used are the same on $\mathbb{T}^3$ with Fourier series, so the statement holds there wherever the endpoint criterion is available. $\square$

The condition is scale-critical and involves half of the degrees of freedom. It is elementary, and we would not be surprised to learn that it is known; we did not find it stated. By Proposition 4.1 it gives nothing on the mirror-symmetric class, where $\|u_-\| = \|u_+\|$ in every norm.

The dynamics also works against controlling one sector. In the helical decomposition the nonlinearity splits into four classes of triads [31]. Numerical experiments find that triads of a single helicity carry an inverse cascade, hidden inside the forward one [2]; that energy is preferentially transferred to the minority helical modes [25]; and that the forward cascade is carried by the mixed classes. A bound on the minority sector would have to run against this transfer.

6. THE GEOMETRY OF BLOWUP: PLANAR STRETCHING

The identity

$$\frac{d}{dt} \|S\|_{L^2}^2 = -2\nu \|S\|_{H^1}^2 - 4 \int \det S \, dx \quad (4)$$

was proved by Neustupa and Penel and rederived through the strain equation by Miller (see [19] for the history); the analogous inviscid statement is due to Chae. It rests on the kinematic identity $\int \operatorname{tr}(\nabla u)^3 = 0$ for divergence-free u . It is local, and $\det S$ is a true scalar, so the quantities built from it satisfy C7. Since $\operatorname{tr} S = 0$, writing $\lambda_1 \leq \lambda_2 \leq \lambda_3$ for the eigenvalues of S , one has $-\det S = \lambda_2(-\lambda_1\lambda_3)$ with $-\lambda_1\lambda_3 \geq 0$, so the cubic term is positive only where $\lambda_2 > 0$. This yields the criterion

$$T^* < \infty \implies \int_0^{T^*} \|\lambda_2^+\|_{L^q}^p \, dt = \infty, \quad \frac{2}{p} + \frac{3}{q} = 2, \quad \frac{3}{2} < q \leq \infty,$$

of [19]: blowup must be driven by two stretching directions and one strongly compressing direction, so by the formation of sheets.

Observation 6.1. Programs that share no methods with (4) single out structures of the same anisotropic kind.

- The blowup of [6], the first proved for the Euler equations from smooth data, occurs at a stagnation ring on the boundary of a cylinder, with swirl odd in the axial variable. There the local Jacobi equation admits no curve-shortening variations and the curvature tends to $-\infty$, the negative-curvature branch of Preston's dichotomy [23], in the form proved for this configuration in [24, Thm. 1.5]. At the ring the velocity gradient has the form $a(e_r e_r^T - e_z e_z^T) + b e_\theta e_z^T$ with $a > 0$, whose symmetric part has eigenvalues a and $\frac{1}{2}(-a \pm \sqrt{a^2 + b^2})$: two positive and one negative, the configuration that the λ_2 criterion singles out (the flow there is inviscid, where the analogue of (4) is Chae's identity without the dissipative term).
- For stationary solutions with finite Dirichlet energy, Coiculescu and Yang show that a nontrivial solution must contain arbitrarily elongated coherent jets [8].
- In the two-and-a-half-dimensional reduction of the $|k|^{2s}$ family studied in [34], the obstruction to regularity is folding, not linear stirring.

None of these implies another. We record only that they point to the same kind of object: coherent, strongly anisotropic structures, and, where the strain can be computed, planar stretching against axial compression.

7. THE TRANSPORT CONDITION

The strain satisfies a closed evolution equation on the space L_{st}^2 of strains of divergence-free fields [19]. Miller's one-parameter family [21],

$$\partial_t S - \Delta S - \frac{1}{2} P_{st}(\omega \otimes \omega) + \mu P_{st}(S^2 + \frac{3}{4} \omega \otimes \omega) = 0, \quad (5)$$

consists of equations on the same constraint space with the same scaling and the same identity (4); here ω is determined by S . At $\mu = 0$ the equation is globally regular, by the identity $\langle -\Delta S, \omega \otimes \omega \rangle = 0$ [21]. At $\mu = \frac{2}{3}$ it is the strain self-amplification model, which blows up in finite time from a large set of data, with enstrophy bounded below by $E_0(1 - r_0 t)^{-2}$ [20]. At $\mu = 1$ it is (1) with the advection term removed, which Miller expects to be at least as singular as $\mu = \frac{2}{3}$.

Two conclusions follow, the first a theorem and the second an inference.

- (1) An argument using only (4), the constraint space and the scaling applies at $\mu = \frac{2}{3}$ and is refuted. Hence neither C2 nor regularity can follow from these ingredients.
- (2) Passing from $\mu = \frac{2}{3}$ to $\mu = 1$ restores the part $\frac{1}{3} P_{st}(S^2 + \frac{3}{4} \omega \otimes \omega)$ of the nonlinearity, which is orthogonal to S and so invisible to (4); Miller argues that this should make the equation more singular, not less [21]. If so, the true equation without advection is singular, and the regularizing mechanism, if the true equation is regular, must be advection. This is a heuristic, not a theorem. It agrees with the numerical evidence of Hou and Lei that removing convection from axisymmetric flows produces blowup while the full equations do not [14].

This is the content of C8. It connects with the structural observation of [33]: the transport of curl m is the same for every inertia operator on the volume-preserving group, and all dependence on the equation sits in the constitutive law recovering u . Condition C8 says the regularizing mechanism, if there is one, lives in the coupling of the two: advection by the exact Biot–Savart velocity, acting against planar stretching. No family of models varies between regularity and blowup continuously enough to transfer a conclusion from one parameter to another; as with the order parameter of [33], there is no interpolation along μ .

8. THE SAME GAP AT INFINITY

The Liouville approach turns the question into one about infinity, and there the missing estimate takes a single concrete form.

Observation 8.1. The following open problems each reduce to controlling the growth, at large scales, of an antiderivative of the velocity.

- *Axisymmetric ancient solutions.* Bounded ancient axisymmetric solutions with $\Gamma \rightarrow 0$ at infinity are constant [17]; the conjecture of [15] is proved when the solution is periodic in the axial variable, where the meridional stream function has bounded oscillation [16]. The open case is a slowly decaying swirl, where that oscillation is uncontrolled.
- *D-solutions.* Stationary solutions on $\mathbb{R}^3$ with finite Dirichlet energy vanish if the stream function has bounded mean oscillation [26], and, by the capsule method, if its mean oscillation on balls of radius R grows like R^α with $\alpha < \frac{1}{9}$ [8]. The energy gives $\alpha = \frac{1}{2}$.
- *Divergence-free drifts.* Incompressibility of a drift cannot by itself break scaling; the good estimates require the drift to be the divergence of a bounded potential [28].

For D-solutions the gap has a simple arithmetic form. Testing the stationary equation with u leaves a boundary flux of order $|u|^3 R^{d-1}$ on spheres of radius R , which vanishes if $u = o(R^{-(d-1)/3})$, while finite Dirichlet energy gives decay of order $R^{-(d-2)/2}$. In $d = 3$ these are

$\frac{2}{3}$ and $\frac{1}{2}$. In $d = 2$ the energy gives no decay but the scalar vorticity obeys a maximum principle; in $d \geq 4$ the energy wins. The three-dimensional problem is the only one in which the energy is short and no scalar maximum principle is available, which is the same stretching-driven split as in [33].

9. THE CANDIDATES AGAINST THE LIST

Table 1 records how the quantities discussed above fare.

	C1	C2	C3	C4	C5	C6	C7	C8
Energy $\ u\ _{L^2}$	no	yes	no	no	–	no	yes	no
Helicity H	yes	partial	no	yes	yes	yes	no	no
λ_2^+ , $\det S$	yes	no	yes	?	yes	yes	yes	no
Swirl $\Gamma = ru^\theta$	yes	special	partial	–	yes	yes	yes	yes
Head pressure Q	yes	special	no	–	yes	yes	yes	yes
Sparseness scale [13]	yes	energy level	yes	?	yes	yes	yes	no

TABLE 1. Candidates and conditions. “Special”: only in the axisymmetric or the stationary class. “Partial”: bounded Γ controls regularity only under a further logarithmic decay near the axis. “–”: not applicable; “?”: not determined here. Helicity is bounded by its initial value only in the inviscid problem; with viscosity (3) has no sign. The a priori content of the sparseness framework is at the level of the energy [1].

The pattern is visible in the table. The quantities that satisfy C2 are integral invariants, and they fail C1 or C7. The quantities that are critical and parity-even are geometric: directions, eigenvalue ratios, scales of sparseness. None of those is bounded a priori. The two rows that come closest, Γ and Q , are scale-invariant scalars with a maximum principle, but only in the axisymmetric and the stationary class respectively, and even there neither controls a critical norm without further assumptions.

Question 9.1. *Is there, for general three-dimensional solutions of (1), a parity-even, scale-invariant scalar quantity satisfying a maximum principle or a monotonicity formula, whose control uses the transport term?*

An affirmative answer would satisfy all eight conditions, and a negative answer in any precise form would explain why the problem has resisted every approach listed here.

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REFERENCES

- [1] D. Albritton and Z. Bradshaw, *Remarks on sparseness and regularity of Navier–Stokes solutions*, Nonlinearity **35** (2022). [arXiv:2110.02187](#).
- [2] A. Alexakis, *Helically decomposed turbulence*, J. Fluid Mech. (2017). [arXiv:1606.02540](#).
- [3] J. T. Beale, T. Kato and A. Majda, *Remarks on the breakdown of smooth solutions for the 3-D Euler equations*, Comm. Math. Phys. **94** (1984), 61–66.
- [4] L. Biferale and E. S. Titi, *On the global regularity of a helical-decimated version of the 3D Navier–Stokes equations*, J. Stat. Phys. **151** (2013), 1089–1098. [arXiv:1303.1215](#).
- [5] D. Chae, *Liouville-type theorem for the forced Euler equations and the Navier–Stokes equations*, Comm. Math. Phys. **326** (2014).
- [6] J. Chen and T. Y. Hou, *Stable nearly self-similar blowup of the 2D Boussinesq and 3D Euler equations with smooth data*, I: [arXiv:2210.07191](#); II: [arXiv:2305.05660](#).
- [7] C.-C. Chen, R. M. Strain, H.-T. Yau and T.-P. Tsai, *Lower bound on the blow-up rate of the axisymmetric Navier–Stokes equations*, Int. Math. Res. Not. (2008); II, Comm. Partial Differential Equations **34** (2009).
- [8] M. P. Coiculescu and J. Yang, *Conditional Liouville theorems for the Navier–Stokes equations*, [arXiv:2506.14533](#).
- [9] P. Constantin and A. Majda, *The Beltrami spectrum for incompressible fluid flows*, Comm. Math. Phys. **115** (1988), 435–456.
- [10] T. M. Elgindi, *Finite-time singularity formation for $C^{1,\alpha}$ solutions to the incompressible Euler equations on $\mathbb{R}^3$* , Ann. of Math. **194** (2021).
- [11] L. Escauriaza, G. Seregin and V. Šverák, *$L_{3,\infty}$ -solutions of the Navier–Stokes equations and backward uniqueness*, Russian Math. Surveys **58** (2003).
- [12] H. Fujita and T. Kato, *On the Navier–Stokes initial value problem I*, Arch. Rational Mech. Anal. **16** (1964), 269–315.
- [13] Z. Grujić and L. Xu, *Asymptotic criticality of the Navier–Stokes regularity problem*, J. Math. Fluid Mech. **26** (2024). [arXiv:1911.00974](#).
- [14] T. Y. Hou and Z. Lei, *On the stabilizing effect of convection in three-dimensional incompressible flows*, Comm. Pure Appl. Math. **62** (2009).
- [15] G. Koch, N. Nadirashvili, G. Seregin and V. Šverák, *Liouville theorems for the Navier–Stokes equations and applications*, Acta Math. **203** (2009). [arXiv:0709.3599](#).
- [16] Z. Lei, X. Ren and Q. S. Zhang, *A Liouville theorem for axi-symmetric Navier–Stokes equations on $\mathbb{R}^2 \times \mathbb{T}^1$* , [arXiv:1911.01571](#); see also [arXiv:1902.11229](#).
- [17] Z. Lei, Q. S. Zhang and N. Zhao, *Improved Liouville theorems for axially symmetric Navier–Stokes equations*, Sci. China Math. (2017). [arXiv:1701.00868](#).
- [18] E. Miller, *Finite-time blowup for the Fourier-restricted Euler and hypodissipative Navier–Stokes model equations*, [arXiv:2307.03434](#).
- [19] E. Miller, *A regularity criterion for the Navier–Stokes equation involving only the middle eigenvalue of the strain tensor*, Arch. Ration. Mech. Anal. (2020). [arXiv:1710.05569](#).
- [20] E. Miller, *Finite-time blowup for a Navier–Stokes model equation for the self-amplification of strain*, Anal. PDE **16** (2023). [arXiv:1910.05415](#).
- [21] E. Miller, *On the interaction of strain and vorticity for solutions of the Navier–Stokes equation*, [arXiv:2407.02691](#).
- [22] OpenAI, *Finite time blowup for the Euler equation*, preprint (2026), not independently verified at the time of writing.
- [23] S. C. Preston, *A geometric rigidity theorem for hydrodynamical blowup*, Comm. Partial Differential Equations **35** (2010), 2007–2020.
- [24] S. C. Preston and A. Sarria, *Lagrangian aspects of the axisymmetric Euler equation*, [arXiv:1505.05569](#).
- [25] G. Sahoo and L. Biferale, *Disentangling the triadic interactions in Navier–Stokes equations*, Eur. Phys. J. E **38** (2015), 114. [arXiv:1510.09006](#).
- [26] G. Seregin, *Liouville type theorem for stationary Navier–Stokes equations*, Nonlinearity **29** (2016).
- [27] J. Serrin, *On the interior regularity of weak solutions of the Navier–Stokes equations*, Arch. Rational Mech. Anal. **9** (1962), 187–195.
- [28] G. Seregin, L. Silvestre, V. Šverák and A. Zlatoš, *On divergence-free drifts*, J. Differential Equations **252** (2012). [arXiv:1010.6025](#).
- [29] T. Tao, *Finite time blowup for an averaged three-dimensional Navier–Stokes equation*, J. Amer. Math. Soc. **29** (2016), 601–674. [arXiv:1402.0290](#).

- [30] T. Tao, *Finite time blowup for Lagrangian modifications of the three-dimensional Euler equation*, Ann. PDE **2** (2016). [arXiv:1606.08481](#).
- [31] F. Waleffe, *The nature of triad interactions in homogeneous turbulence*, Phys. Fluids A **4** (1992), 350–363.
- [32] A. Salkimbayev, *Momentum helicity separates cascade operators from Euler–Arnold operators*, companion paper.
- [33] A. Salkimbayev, *A Beale–Kato–Majda criterion for right-invariant H^s metrics on $\text{SDiff}(\mathbb{T}^3)$* , companion paper.
- [34] A. Salkimbayev, *Two-and-a-half-dimensional flows of the H^s Euler–Arnold equations: a self-gravitating Boussinesq system*, companion paper.

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